EVALUATION OF NLP SYSTEMS

Lecture 3 - Common procedures

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- measurements
- baseline
- dataset partition
- cross validation
- error analysis
- ablation study
- human evaluation
- comparing

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EXPERIMENTING

EVALUATING

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Measurements

can be created, adapted or customized according to tasks

quantify the correctness or the quality of system output

metrics
scores
scales
statistics
measures

need context to be meaninful

usually conventions: do not trust them blindly

off-the-shelf implementation in many libraries (e.g. sklearn.metrics for Python)

details in upcoming lecture

Baseline

two main purposes

a trivial system as sanity-check

if a fancy algorithm cannot beat outputs generated randomly, something is wrong

an existing system or previous version as lower bound

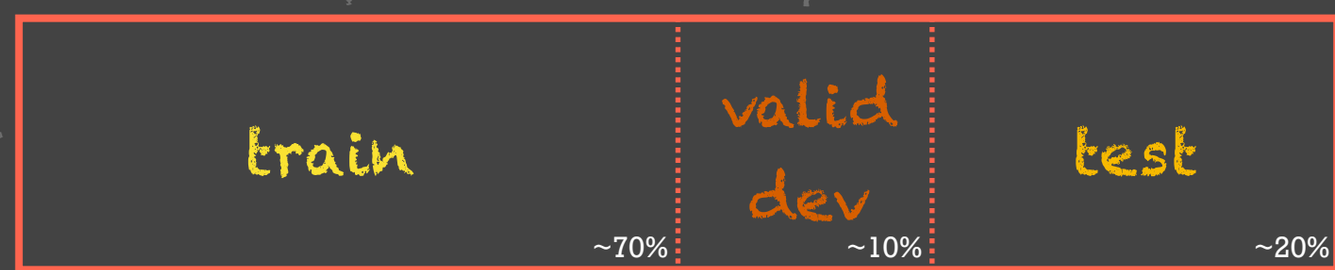
check whether a new approach has any benefit over an already good one

Dataset partition

pretend this is your test set until you are ready to test for real

data your algorithm sees during the learning process

use to check performance on (really) unseen data



beware of overfitting

use for hyperparameter tuning and checking learning progress in unseen data

don't ever use this until you are ready to test (once!) and report results

Cross validation

when we don't have

- much data
- a fixed train-valid-test split

⇨ create partitions

⇩

average results

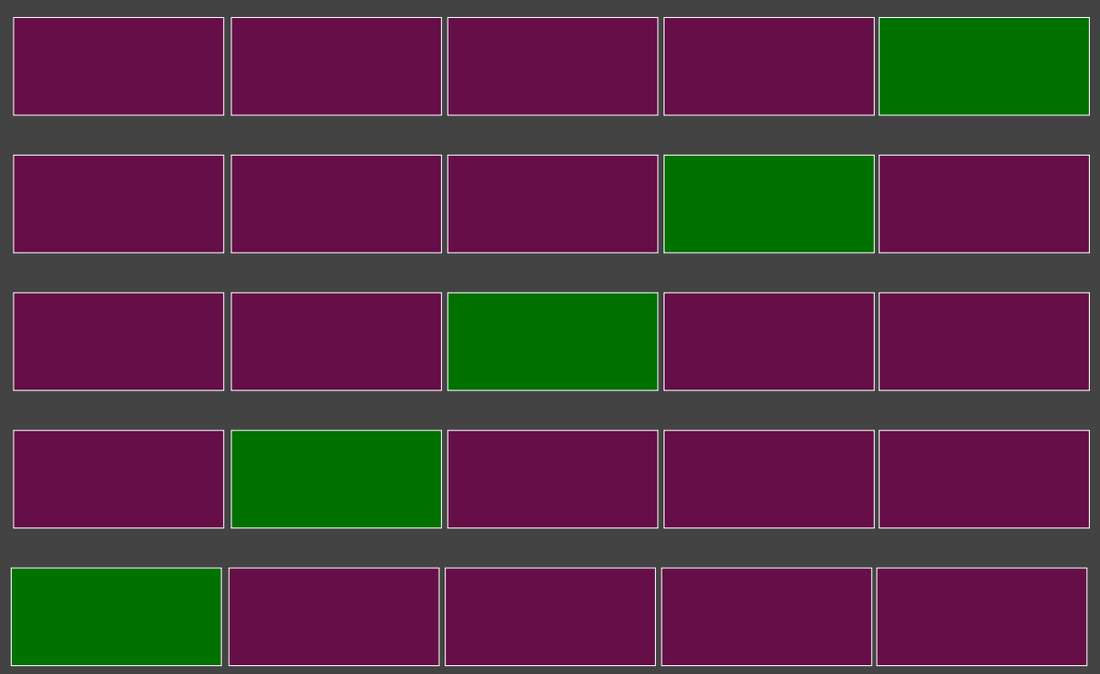
Common methods

- ✓ holdout
- ✓ random splits
- ✓ k-fold
- ✓ leave-one-out
- ✓ leave-p-out

Example: 5-fold

DATA

partition it into 5 randomly sampled, equal-sized folds



run 5 times (keeping folds fixed)

training with all

testing with

get 5 estimates

r_1

r_2

r_3

r_4

r_5

mean

$$\frac{r_1 + r_2 + r_3 + r_4 + r_5}{5} = R$$

our final estimate

Error analysis

know your data

look at your outputs

what is the system doing wrong

when, how, why errors occur

identify patterns and causes

classify them

adapt system

errors propagate!
check how downstream
tasks are affected

do it on dev
(not test) set

Ablation study

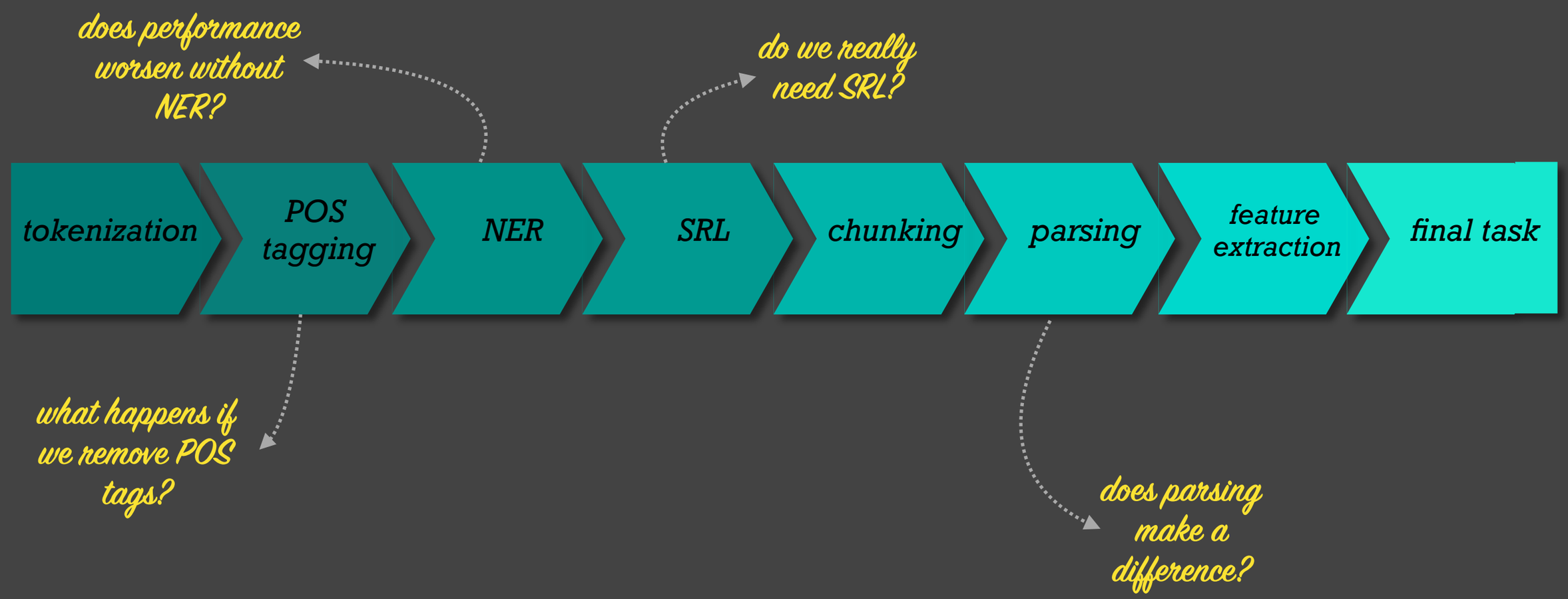
what's the contribution of parts to the whole?

when the total system is available but uncertainty exists about the relation of anatomy to functioning

remove various components and study the performance of the diminished system

see how well the surviving mutilated system performs in a good tactic of performance analysis

Newell (1975)



Human evaluation

- subjetive judgements
- possibly biased
- sometimes needs experts
- unfeasible to repeat many times
- costly + time consuming

PROS AND CONS

a mix of automatic evaluation and human judgement is usually welcome

- better to capture subjective criteria (fluency, naturalness, quality etc)
- check preferences between outputs
- in line with main purpose (satisfied users)
- gold standard annotation
- easy to crowdsource nowadays

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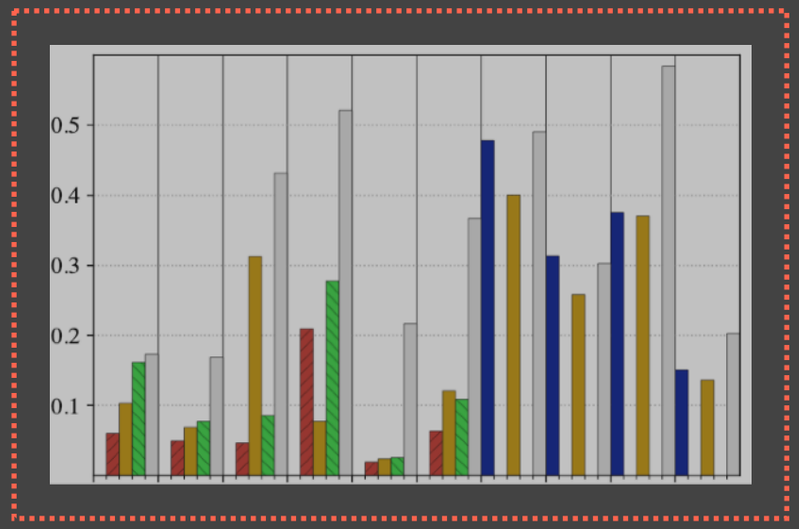
Comparing performance

☞ *use tables and plots to summarize results and compare systems*

Table 11.1 Structure of a typical summary of evaluation results

Condition	*Measure 1*	*Measure 2*	*Combined Measure*
Baseline 1	M_1^{B1}	M_2^{B1}	M_c^{B1}
Baseline 2	M_1^{B2}	M_2^{B2}	M_c^{B2}
Variation 1	M_1^{V1}	M_2^{V1}	M_c^{V1}
Variation 2	M_1^{V2}	M_2^{V2}	M_c^{V2}
Upper Bound	M_1^{U}	M_2^{U}	M_c^{U}

from Resnik & Lin (2010)



better measures alone are no guarantee of overall superior performance of a system

maybe it was lucky only in this particular dataset… but what if we don't have other datasets?

statistical significance tests

details in upcoming lecture

See you next week!

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